DAY 22 · THEORY HANDOUT

Join Steps Into a Chain

Use LCEL to pass a prompt through a model and a text parser.

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The idea and its useful terms

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Use LCEL to pass a prompt through a model and a text parser.

Runnable

A step we can call with an input value.

LCEL

LangChain Expression Language joins steps.

Pipe operator

The | symbol passes one step's output onward.

A chain records the order of work. We can run the whole path again with new input instead of rebuilding it by hand.

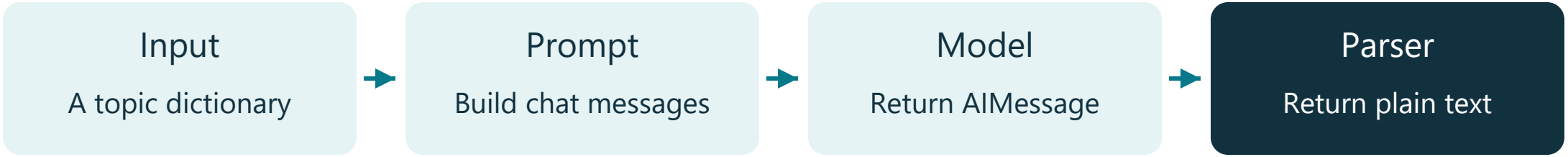
Each step expects a certain input type. A parser converts the model reply into the type the next step needs.

> A second model call is not a fact checker. Check important facts against the supplied input.

Follow the work, one step at a time

Each step has a job. We can inspect its output before moving on.

Read the path from left to right.



Each arrow means: pass the result to the next step.

The first step accepts a dictionary and returns a prompt value. The chat model accepts that prompt and returns AIMessage. StrOutputParser accepts the reply and returns a string. These types explain the order.

A second chain can accept the first string as its input. Show that intermediate string before asking for a shorter version. If the first reply is wrong, a shorter version may repeat the same error.

> If the answer is wrong, inspect the input and the intermediate results. The last sentence alone cannot show which step failed.

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A small example we can check

The values here are classroom examples. The notebook shows actual results.

Step 1

topic = solar system

A dictionary enters.

Step 2

The model explains it.

An AIMessage leaves.

Step 3

StrOutputParser reads it.

A string leaves.

Read the given input first. Point to the fact or value needed for the requested output.

Run the matching example, then try the fresh input at the end of the notebook. Compare both results with the known input.

> A second model call is not a fact checker. Check important facts against the supplied input.

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Choose the setting or tool for its job

A different choice changes the work. These are the choices we compare.

invoke

Run one input.

Good for an early check.

batch

Run a list of inputs.

More calls, more work.

Two-stage chain

Explain, then shorten.

A second call can add errors.

Build prompt | model | StrOutputParser and display its string output.

Use a second prompt to shorten the first answer. Show the intermediate text.

> Repeat with a new topic. Check whether the shorter text still keeps the same facts.

Practice, check, and read the full lesson

Use the notebook to try the steps. Use the article when you need the longer explanation.

1. Build

Run the small example.

Inspect each step.

2. Change

Try the fresh input.

Keep the goal clear.

3. Check

Compare with the source.

Keep mistakes visible.

Repeat with a new topic. Check whether the shorter text still keeps the same facts.

Read the complete tutorial breakdown for this section in our KGP Talkie article.

> A second model call is not a fact checker. Check important facts against the supplied input.

DAY 22 · COMPLETE

Thank You

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